Street-Level Geolocation Across 12 European Countries

Progress report · generated 2026-08-29 · branch `pini/uni_machine_code`

1. Problem statement

Given a 512×512 street-level photograph taken in one of **12 European countries** (Belarus, Finland, France, Germany, Iceland, Italy, Norway, Poland, Spain, Sweden, Turkey, United Kingdom), predict the GPS coordinate `(lat, lng)` at which it was taken.

Item	Detail
Training data	11,758 labelled images (`filename, country, iso, lat, lng`), 864–1000 per country
Test data	2,400 unlabelled "holdout" images — we submit `predictions.csv`
Metric	**Median haversine distance (km)** to the true coordinate over the 2,400 test images. Lower is better. Mean distance is the tie-break. Reported secondary rates: <200 km ("got the city"), <750 km ("got the country").
Hard constraints	(1) submitted model ≤ **5,000,000 parameters** (pretrained weights count); (2) train **only** on the provided images — no external datasets, no scraped or generated data.
Grading	50% method & code quality · 35% write-up · 15% leaderboard

2. Where we stand

139.8 km
current *submitted* holdout score (rank #5 / 12)

122.3 km
best cross-validation median achieved this session (fold 0)

47.4 km
leaderboard #1 (the target to chase)

A better submission file is ready but **not yet uploaded**:
`submission_candidates/preds_kdtree_fulldata_mode.csv` — from the current best single model trained on 100% of the data. Expected holdout ≈ 125–130 km (→ rank #4).

The competition leaderboard (as observed)

Rank	Participant	Median km	Mean km	<200 km	<750 km	Subs	Params
1	Vaibhav Prajapati	47.4	372.5	65.4%	82.3%	11	4.52M
2	Kamaal Ahmed	62.7	367.7	63.4%	84.2%	3	4.90M
3	Joswin Dsouza	85.5	417.7	61.5%	81.5%	3	4.44M
4	Aleksei Bezgin	118.7	439.5	58.1%	80.1%	1	—
5	**Us (Praveen N. Hegde)**	**139.8**	438.5	55.4%	80.2%	4	4.72M
6	Jasmeet Singh	188.0	645.6	51.0%	71.3%	2	4.71M

What the leaderboard reveals. The `<750 km` (correct-country) rate is **~82% for every rank, top to bottom** — nobody, including #1, is meaningfully better at country classification, and the means are all ~370–440 km,

so nobody fixes the catastrophic tail. The *entire* ladder from #5 to #1 is the `<200 km` (correct-city) rate: 55% → 65%. The gap is **within-country precision** on the ~82% of images whose country we already get right.

3. The model architecture

A retrieval-augmented hierarchical classifier, evolved over the project. Current form:

- **Backbone**: ImageNet-pretrained EfficientNet-B0 (`efficientnet_b0.ra4_e3600_r224_in1k`, via `timm`). ~4.0M of the 5.0M parameter budget. Not frozen (all params counted).
- **Multi-view input**: 5 crops per image (1 global + 4 corner detail crops) through the *shared* backbone, then fused.
- **Heads** on the fused feature: (a) 12-way country classifier; (b) **K = 4 independent geo-cell classifiers** — four separately-seeded recursive partitions of the training points, each cell belonging to exactly one country ("CPlaNet" combinatorial partitioning); (c) a within-cell offset regressor; (d) a 224-d L2-normalised embedding.
- **Inference decode**: score *every labelled training image* by $\log P(\text{country}) + \text{mean}_h \log P(\text{cell}_k \mid \text{country}) + w \cdot \text{cosine-similarity}(\text{embedding})$, then output a density-weighted spherical mean over the strongest coherent geographic cluster among the top candidates (mode-seeking retrieval).
- **Training losses**: haversine-smoothed cell cross-entropy + country CE + offset Huber + a geo-distance-matching loss on the embedding + a 5-view consistency loss + a supervised geographic contrastive loss.

4. Methods tried — full log

All numbers are the honest fixed-decoder **median km on a held-out fold** (group-stratified 5-fold; 500 m location clusters kept whole so near-duplicates never straddle train/val). Fold-0 unless stated.

4.1 Pre-session baseline (the currently-submitted model)

Tag	What	Median	Outcome
T2.1	Direct (lat,lng) regression, MobileViT-XS, low resolution	450–500	Badly underfit. Regression averages between plausible modes.
T3.1	Hierarchical country + single geo-cell classifier, EfficientNet-B0	193	Big jump — classification > regression here.
T3.4	K independent coarse cell partitions, combined at decode (CPlaNet)	183	K=2 helps; further gains to K=3.
T4.1b	Shared-backbone multi-view (global + 4 local) retrieval + mode decode	155	Retrieval against the training gallery.
T4.1c	+ supervised geo-contrastive loss on the embedding	149.5	**5-fold CV**. The recipe of the submitted model.
T5.x	Backbone swaps (FastViT-T8, RepViT, StarNet), 384 px fine-tune, per-country loss up-weighting for Sweden/UK/Turkey	150–158	All matched or lost. Documented as negative results.
T5.3	Final model: T4.1c recipe, trained on 100% of the data	150.6 (CV) → **139.8 (holdout)**	The current submission. Holdout ~11 km better than CV.

4.2 This session (T6) — the push toward the leaderboard

Tag	Change tested (one variable at a time)	Median	<200	cell acc	Outcome
E1	Add the provided `orientation_flags.json` correction (rotate the ~9% of images flagged upside-down) + K=4 cells, retrain from scratch	153.6	.56	.43	Neutral on CV (the CV model also learns to handle flipped images). Kept — near-zero risk, and the flag also covers the holdout where the model can't adapt.
E2	Warm-start E1 + resolution 384 + label smoothing + EMA + region-contrastive (4 changes bundled)	~170	—	.35↓	**Failed.** Bolting label smoothing onto an already-trained model collapses the cell classifier. Lesson: never bundle; smoothing must be from-scratch.
E2b	Warm-start E1 + resolution 384 *only*	163–169	—	—	**Negative.** Resolution is not the lever at EfficientNet-B0 scale.
E3c	Finer cells 96 → 192 per partition, plain one-hot loss	~160	—	.34	**A wash.** Precision gain of smaller cells is exactly cancelled by the accuracy loss (192-way classifier tops out at 34%).
E3d	E3c + **haversine label smoothing** (PIGEON): soft cell target = softmax(−distance/τ) over that image's country's cells, τ≈65 km	**148.5**	.57	.23	**First real gain.** Neighbouring cells share the label mass → classifier rewarded for landing in the right *area*. Cell top-1 drops but the *distribution* is well-placed.
E3e	E3d taken further: **368 cells** per partition + a shared low-rank trunk before the cell heads (to fit budget) + a 55-epoch schedule + τ≈45 km	**137.2** (132.7 swept)	.59	.13	**Best single model.** Decoder sweep (higher retrieval weight, 150 km cluster radius) → 132.7.
E5 (f1,f2)	E3e recipe on CV folds 1 and 2	135.3 / 138.0	.57/.58	.13	Confirms the recipe generalises — fold medians 137.2 / 135.3 / 138.0 are near-identical.
Ens.	Score-space fusion of 4 diverse models (E3e + a GeoCLIP variant + E3d + E3h)	**122.3**	.62	—	**Best result achieved.** Mean drops 428 → 368: a majority of diverse models rarely all misplace the same image. (Note: the challenge asks for a *single* model — see §6.)
E3f	E3e + 2.5× country-loss weight + region-level contrastive positives	worse	—	—	**No-op on country accuracy** (0.744, identical to E3e), median regressed. The ~78% country ceiling is a capacity/feature limit, not a loss-weight one.
E3g	E3e + MLP country head + resolution 384	stalled	—	—	Country accuracy went *backwards*. Killed at epoch 8.
—	Country-gated decode: hard-mask retrieval to the top-k predicted countries; boost the country prior	no change	—	—	France stays at 467 km even with a top-2 country mask — when the model is wrong about France it is *confidently* wrong (France not in its top 2).
E3h	E3e + tighter local crops (64% → 40% width)	~neutral	—	—	Detail crops slightly more informative, but the 5-view

					consistency loss doubled and cancelled it.
E4	GeoCLIP-style continuous location encoder (equal-earth + random Fourier features + MLP) sharing the embedding, InfoNCE image↔location loss	worse standalone	—	—	The auxiliary alignment loss drags the retrieval embedding. Useful only as an ensemble member.
—	Sharp "nearest-training-image" decode (the spatial-overlap exploit hypothesis)	worse	—	—	Analysis showed the holdout's retrieval-similarity distribution is *identical* to our cluster-separated CV → **the holdout is genuinely spatially separated**, there is no near-duplicate exploit, and CV is a faithful proxy for the holdout.
—	Within-cell offset regression + confidence-adaptive decode (OSV-5M hybrid)	no gain	—	—	The Huber offset head learned nothing (cell+offset 172 vs centroid 169). Mode-retrieval beats cell-centroid at *every* confidence level. The decode is not the bottleneck.
E7	**k-means cells** instead of kd-tree (per-country MiniBatchKMeans; compact blobs, not axis-aligned strips)	running	—	—	Oracle ceiling 33 → **25 km** at the same cell count. In progress.
E8	k-means cells + **country → region → cell hierarchy head** (nested easy decisions)	queued	—	—	Wired & tested (+62k params, fits). In queue after E7.

5. Diagnosis — what is limiting us

5.1 The gap is within-country precision, not country classification

Per-country median error of the best (122 km) ensemble, fold 0:

Country	n	median km		Country	n	median km
Iceland	192	59		United Kingdom	200	117
Finland	200	74		Poland	187	206
Belarus	173	79		Spain	200	216
Norway	200	86		Germany	200	243
Turkey	200	99		Italy	200	258
Sweden	200	103		**France**	200	**373**

The overall median sits **exactly at the boundary** between the 7 "easy" countries (all < 120 km) and the 5 "hard" continental ones. Crucially, for the easy countries the model already *beats* the "know the country perfectly" oracle (e.g. Iceland 59 km vs a 178 km country-centroid oracle) — the cell system works well when the country is right. For France the model is *worse* than the country-centroid oracle (373 vs 330): it is predicting the wrong country outright (~42% of French images land > 500 km, mostly labelled Germany/Italy/Poland).

5.2 The cell classifier collapses at fine granularity

cells / partition	oracle median (K=4, perfect classification)	actual cell top-1 accuracy
96	84.5 km	43%
192	54.1 km	34%
368	33.1 km	13%

Every time we halve the cell, the *best possible* result improves, but the classifier gets the cell right far less often (fewer training images per class, a harder problem). The haversine label smoothing recovers most of the *median* (because the soft distribution is still roughly in the right place) but not the top-1 accuracy — so **few predictions land inside a tight cell, hence few `<200 km` hits.** This is the mechanism behind our 59% vs #1's 65%.

5.3 The hard core: featureless rural continental Europe

The error mass is ~500 validation images where the cell classifier has almost no confidence (top probability ~15%). These are generic rural roads in France / Germany / Poland where a single street-level frame genuinely looks the same across borders. For these images only the embedding provides signal, and the geo-distance / contrastive losses have little to grab onto in a featureless scene. This is a shared ceiling — the whole leaderboard is at ~82% correct-country — but #1 clearly extracts *some* extra signal (road markings, utility poles, vegetation, sky/light) that our model at 288 px is not learning.

5.4 The 5,000,000-parameter budget

EfficientNet-B0's trunk alone is ~4.0M, leaving only ~1M for all the classification heads — and classification capacity is exactly the bottleneck. A smaller backbone with much larger heads (or a genuinely stronger sub-5M feature extractor) is an open, higher-risk lever.

6. Constraints and lessons

- **Single model required.** The clarifications say "submit a single final model", and every leaderboard entry is one model ≤ 5M params. Our best result (122 km) is a 4-model ensemble, likely non-compliant. Mitigation in progress: a *model soup* (weight-average of same-architecture seeds) — one model, ensemble-like robustness.
- **The holdout is spatially separated** from training (verified via retrieval-similarity distributions). No nearest-neighbour shortcut. CV is a faithful proxy.
- **Cross-validation is trustworthy**: fold medians 137.2 / 135.3 / 138.0 for the same recipe; old submission CV 150.6 → holdout 139.8.
- **Process lessons**: never bundle >1 change per screen; label smoothing must be trained from scratch (not fine-tuned in); EMA decay ≥ 0.999 makes early evaluations meaningless; the smoothing temperature τ must be scaled to the inter-cell spacing.

7. Current plan to close the gap

Three compounding levers, all aimed at **cell accuracy at fine granularity** (the §5.2 problem):

1. **Coherent k-means cells** (E7, running) — compact blobs classify better than kd-tree strips; oracle 33 → 25 km.
2. **Nested country → region → cell hierarchy** (E8, queued) — three easy decisions instead of one impossible 1-of-368; a wrong cell still lands in the right region.

3. **Test-time augmentation** (flip + multi-scale + native-resolution corner crops) and a longer / EMA-tuned schedule — reliable few-km gains, cheap.

Realistic projection: 122 → ~95–105 km (roughly rank #3). Reaching #1's 47 km likely also needs semantic cells and a training regime found through many submission iterations, and possibly a different backbone.

> **Immediate action available:** upload `submission_candidates/preds_kdtree_fulldata_mode.csv` — the best current single model, trained on 100% of the data, mode decode + horizontal-flip TTA. Expected holdout ≈ 125–130 km, moving us from rank #5 to #4 while the E7/E8 work continues.

Report reflects state at 2026-08-29 20:45. Experiment logs, per-experiment READMEs (including every failed run) and the full cell-accuracy oracle analysis are in `outputs/experiments/`, `failed_experiments/` and `T6_FINDINGS.md` in the repository.